HYPERFINITENESS OF A COUNTABLE BOREL EQUIVALENCE RELATION ON A POSITIVE RAMSEY CONE

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Editorial note. The author explicitly assumes metric closedness for the Ramsey space at470, so the counted statement is read with that source-global convention. Only the listed classical Ramsey-space properties are free inputs; the separated-cover criterion and fusion scheduling are proved.

Editorial note. In641 the indices of theB-set interval are shifted by one stage. The construction638-639 assigns the cones at stage n to k_n<k<=k_(n+1). A point C in[A] is below A_(n+1), so its relevant prefix cone lies in the B_k selected at that same stage. The original641 is preserved and this bookkeeping interpretation requires independent confirmation.

Editorial note. In the diameter argument602-605, a component with no current marker is unchanged. A component with one marker consists of its old component together with the old components joined by the new edges of G_(n+1) restricted to B; all paths then have length at most2(f(n)+1). The displayed union604 is read within that restricted vertex set.

Editorial note. The later generic-comeager theorem664-676 is not used or counted here; no inference about saturation of a meager set is needed for this task.

Editorial scope. One arbitrary-topological-Ramsey-space hyperfiniteness theorem, with the full separated-cover and fusion proof chains. Its Ellentuck special case and the old bounded-degree lemma are not counted or used as a shortcut. Standard Borel equivalence and abstract Ramsey facts are declared precisely. Source index/union conventions remain explicitly subject to independent review.

Allowed standard prerequisites (editorial).

- Classical topological Ramsey-space fusion and Ramsey properties. Use the metric-closed topological Ramsey-space setting stipulated in source470. Basic cones[a,A] are Borel, each finite downset{s:s<=fin r_n(A)} is finite, fusion sequences have the stated limit, and the two fusion conclusions474-480 and finite Borel-color homogenization484-486 hold.. Todorcevic, Introduction to Ramsey Spaces; exact properties supplied in source449-486.. Provides individual homogeneous refinements and fusion limits, not the separated-cover scheduling or its hyperfinite finite-component construction.
- Classical countable Borel equivalence-relation facts. A CBER is the union of countably many graphs of Borel involutions. Hyperfiniteness is equivalent to being an increasing union of finite Borel equivalence relations, and is inherited by restriction to a Borel subset.. Convert the constructed finite components into hyperfiniteness and represent an arbitrary E by bounded-degree graphs.
- Finite Borel vertex coloring for bounded-degree irreflexive graphs. Such a graph has a finite Borel proper vertex coloring (Kechris-Solecki-Todorcevic).. Source625.. Applied to a bounded graph-distance power before the Ramsey homogeneous refinement.

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A *countable Borel equivalence relation (CBER)* on a Polish space X is an equivalence relation with all classes countable that is Borel as a subset of X^2. CBERs have a rich and well investigated structure theory (see, e.g., [1]). The natural way to compare two such equivalence relations is the notion of *reduction*: if E and F are CBERs on spaces X and Y, E is said to *Borel reduce* to F if there exists a Borel map $\phi : X \to Y$ with

$$\forall x, x' \in X \; xEx' \iff \phi(x)F\phi(x').$$

The simplest non-trivial class of CBERs is the collection of *hyperfinite* ones, that is, the collection of CBERs which admit a Borel reduction to $\mathbb{E}_0$, the eventual equality equivalence relation on $2^{\mathbb{N}}$.

Recall the following definition.

Definition 0.1. Let G be a graph, $B \subseteq V(G)$. The set B is called *k-separated* if for any $x \neq x' \in B$ we have $\mathrm{dist}_G(x, x') > k$.

A *topological Ramsey space* in the sense of Todorčević [3] is a triple $(\mathcal{R}, \leq, r)$ satisfying a certain set of axioms, where $\mathcal{R}$ is a nonempty set, $\leq$ a quasi-ordering on $\mathcal{R}$, and $r : \mathcal{R} \times \mathbb{N} \to \mathcal{AR}$. The range of r can be thought of as the collection of finite approximations to elements of $\mathcal{R}$. We will use capitals $A, B, \dots$ for the elements of $\mathcal{R}$, and $a, b, \dots$ for their approximations. We denote $r_n(.) = r(., n)$.

The relation $\leq$ is encoded on the level of approximations by a quasi-ordering $\leq_{\mathrm{fin}}$ on $\mathcal{AR}$.

For $a \in \mathcal{AR}$ and $B \in \mathcal{R}$ define

$$[a, B] = \{A \in \mathcal{R} : A \leq B \wedge (\exists n)\; r_n(A) = a\}.$$

We also use the abbreviations $[B] = \{A \in \mathcal{R} : A \leq B\}$ and $[n, B] = [r_n(B), B]$.

There are multiple natural topologies on $\mathcal{R}$: the *metric topology* of $\mathcal{R}$ inherited from the Baire space $\mathcal{AR}^{\mathbb{N}}$, and the so-called *Ellentuck topology* generated by $\{[a, A] : \; a \in \mathcal{AR}, \; A \in \mathcal{R}\}$. We work with the metric topology, unless stated otherwise.

The next definition gives a standard way of constructing elements of $\mathcal{R}$ with some desirable property.

Definition 0.2. A *fusion sequence* is a sequence $(A_n)_{n\in\mathbb{N}}$ in $\mathcal{R}$ such that $A_{n+1} \in [n, A_n]$ for every $n \in \mathbb{N}$.

The *limit* of a fusion sequence $(A_n)_{n\in\mathbb{N}}$, denoted $\lim A_n$, is the unique $A \in \mathcal{R}$ such that $A \in [n, A_n]$ for every $n \in \mathbb{N}$.

Note that for the existence of the limit of a fusion sequence we assume that $\mathcal{R}$ is closed in $\mathcal{AR}^{\mathbb{N}}$ with respect to the metric topology. This is not listed among the axioms of topological Ramsey spaces in [3], however, this property is assumed for the main results of the theory. We make use of the following properties of fusion sequences.

Proposition 0.3. *Let (A_n) be a fusion sequence and $C \in [\lim A_n]$. Then*

1. *$\lim A_n \leq A_k$ for every k,*
2. *there are infinitely many n with $C \in [s, A_n]$ for some $s \leq_{fin} r_n(A_n)$.*

Apart from the technique of fusion sequences, we use the Ramsey theoretic statement below.

Theorem 0.4. *Let $\mathcal{R}$ be a topological Ramsey space, $c : \mathcal{R} \to k$ a finite Borel coloring, $A \in \mathcal{R}$, and $s \leq_{fin} r_n(A)$. Then there exists $B \in [n, A]$ such that $c \restriction [s, B]$ is monochromatic.*

Theorem 0.5. *Let $\mathcal{R}$ be a topological Ramsey space and E be a CBER on $\mathcal{R}$. Then there is a set $A \in \mathcal{R}$ such that $E \restriction [A]$ is hyperfinite.*

Theorem 0.6. *Let E be a CBER on the space X, let $(G_n)_{n\in\mathbb{N}}$ be an increasing sequence of bounded degree Borel graphs*[1] *such that $\bigcup_n G_n = E$, and let $f : \mathbb{N} \to \mathbb{N}$ be such that $\forall n \in \mathbb{N}\ f(n+1) \geq 2 \cdot (f(n)+1)$. Moreover, assume that $B_n \subseteq X$ are Borel sets so that every B_n is $f(n)$-separated in G_n.*

Then $E \restriction B$ is hyperfinite, where $B = \{x : \exists^\infty n\ (x \in B_n)\}$.

Proof. We define a sequence $H_0 \subseteq H_1 \subseteq \cdots \subseteq H_n \subseteq \cdots \subseteq E$ inductively. Let $H_0 = \emptyset$.

Given H_n, let

$$H_{n+1} = H_n \cup \{(x,y) \in G_{n+1} \restriction B :\ x \in B_{n+1} \text{ or } y \in B_{n+1}\}.$$

First we check that $\bigcup_{n\in\mathbb{N}} H_n \restriction B = E \restriction B$. Indeed, let $(x,y) \in G_n \restriction B$. Then there is some $k \geq n$ such that $x \in B_k$. But then $(x,y) \in H_k$.

Claim 0.7. *Every connected component of H_n has diameter $\leq f(n)$.*

Proof. We prove this by induction on n. The case $n = 0$ is clear, so assume that we have shown the statement for n. We claim that if H is a connected component of H_{n+1} then it contains at most one point of B_{n+1}. Indeed, otherwise let $x \neq x'$ from B_{n+1} in $V(H)$ be such that for $p = (x_0, \ldots, x_k)$, the injective path with minimal length from $x_0 = x$ to $x_k = x'$ in H we have $x_1, \ldots, x_{k-1} \notin B_{n+1}$. Then $x_1, \ldots, x_{k-1}$ are contained in a single connected component H' of H_n, and by the inductive hypothesis $\mathrm{dist}_{H_n}(x_1, x_{k-1}) \leq f(n)$. But this implies $\mathrm{dist}_{G_{n+1}}(x, x') \leq f(n) + 2 \leq f(n+1)$, a contradiction.

Thus, H contains at most one point, say x, from B_{n+1}. Then

$$V(H) = \bigcup \{[y]_{H_n} : (x,y) \in G_{n+1}\}.$$

yielding

$$\mathrm{diam}(H) \leq 2 \cdot (f(n)+1) \leq f(n+1).$$

□

As $H_n \subseteq G_n$ which has bounded degrees, this, in turn also ends the proof of Theorem 0.6. □

Lemma 0.8. *Let $\mathcal{R}$ be a topological Ramsey space, let G be a bounded degree Borel graph on $\mathcal{R}$, let $k \in \mathbb{N}$, and let $a \in \mathcal{AR}$ and $A \in \mathcal{R}$ be such that $a \leq_{fin} r_n(A)$. Then there exists $A' \in [n, A]$ such that $[a, A']$ is k-separated.*

Proof. By [2], every irreflexive bounded degree Borel graph admits a finite Borel vertex coloring. Let $c : \mathcal{R} \to m$ be such a coloring of

$$G^k = \{(x,y) : x \neq y \wedge \mathrm{dist}(x,y) \leq k\}.$$

By Theorem 0.4. there is some $A' \in [n, A]$ such that $c \restriction [a, A']$ is constant, thus, $[a, A']$ is k-separated. □

Lemma 0.9. *Let $\mathcal{R}$ be a topological Ramsey-space, G_n a sequence of bounded degree Borel graphs on $\mathcal{R}$, and $f : \mathbb{N} \to \mathbb{N}$. Then there exist $A \in \mathcal{R}$ and $B_n \subseteq \mathcal{R}$ Borel sets such that every B_n is $f(n)$-separated in G_n, and*

$$[A] \subseteq \{C : \exists^\infty n\ (C \in B_n)\}.$$

Proof. We construct a fusion sequence $(A_n)_{n\in\mathbb{N}}$ inductively. Let

$$k_n = \sum_{i<n} |\{s \in \mathcal{AR} :\ s \leq_{\mathrm{fin}} r_i(A_i)\}|.$$

Let $B_0 = \emptyset$. Throughout the construction, we define $B_{k_n+1}, \ldots, B_{k_{n+1}}$ after fixing A_n (and k_{n+1}). Suppose that A_n and $B_0, \ldots, B_{k_n}$ are given. We enumerate $\{s \in$

[1] Throughout the paper, graphs are not necessarily assumed to be irreflexive.

$\mathcal{AR}: \ s \leq_{\mathrm{fin}} r_n(A_n)\}$ as $\{s_j : 1 \leq j \leq k_{n+1} - k_n\}$, and define $(A'_j)_{j \leq k_{n+1}-k_n}$ recursively as follows: let $A'_0 = A_n$, and if A'_j is already given for some $j < k_{n+1}-k_n$, choose $A'_{j+1} \in [n, A'_j]$ so that $[s_{j+1}, A'_{j+1}]$ is $f(k_n + j + 1)$-separated in G_{k_n+j+1} (such an element of $\mathcal{R}$ exists by Lemma 0.8), and let $B_{k_n+j+1} = [s_{j+1}, A'_{j+1}]$. Afterwards, let $A_{n+1} = A'_{k_{n+1}-k_n} \in [n, A_n]$.

Finally, let $A = \lim A_n$ and suppose that $C' \in [A]$. By Proposition 0.3. there are infinitely many n with $C' \in [s, A_n]$ for some $s \leq_{fin} r_n(A_n)$. For such an n and s, we have that $[s, A_n] \subseteq B_k$ for some $k_{n-1} < k \leq k_n$, which yields $C' \in \{C : \exists^\infty n \ (C \in B_n)\}$. □

Proof. As all classes of E are countable, there are Borel involutions φ_n on $\mathcal{R}$ such that $E = \bigcup_{n\in\mathbb{N}} \mathrm{graph}(\varphi_n)$. Let $G_n = \bigcup_{i\in n} \mathrm{graph}(\varphi_i)$, and $f : \mathbb{N} \to \mathbb{N}$ be as in Theorem 0.6. By Lemma 0.9. there exist $A \in \mathcal{R}$ and $B_n \subseteq \mathcal{R}$ Borel sets such that every B_n is $f(n)$-separated in G_n, and

$$[A] \subseteq \{C : \exists^\infty n \ (C \in B_n)\},$$

thus, $E \restriction [A]$ is hyperfinite according to Theorem 0.6. □

References

[1] Alexander S Kechris. *The theory of countable Borel equivalence relations*. Cambridge University Press, 2024.

[2] Alexander S. Kechris, Sławomir Solecki, and Stevo Todorčević. Borel chromatic numbers. *Adv. Math.*, 141(1):1–44, 1999.

[3] Stevo Todorcevic. *Introduction to Ramsey spaces*, volume 174 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 2010.